Summer Term Assessment Period 2021

Assessment paper and instructions to candidates:
MY457 – Causal Inference for Observational and Experimental Studies

Suitable for all candidates

Instructions to candidates

This paper contains **3** questions, each with various sub-questions. Answer **all** questions and sub-questions. Mark allocation is as follows:

Question 1: 32 marks
Question 2: 26 marks
Question 3: 42 marks

Marks for each sub-question are indicated in brackets in the exam.

Answers should be justified by showing work wherever appropriate.

Time allowed to complete assessment: 3 hours

Additional time added for assessment upload: 30 minutes

This is an "open book" online assessment. You are free to use any written material you find useful. You are also allowed to use any type of calculator.

Further instructions:

- You should download the assessment paper and submit your answers through the online assessment link on the MY457 Moodle page.
- Please see the submission coversheet for information about how you should submit your answers (this is the other document you downloaded from the Moodle page, in addition to this assessment paper).
- The work you submit is expected to be 100% your own. Therefore, you must not collaborate or confer with anyone during the assessment. The School may carry out checks to ensure the academic integrity of your work.
- You are not required to include a bibliography or a list of references at the end of your submission. Whenever you use the words or thoughts of someone else directly, you need only reference it in (Name, Year) format in the text of your answer. When inserting a section of text (of any size) from someone else's work into your own, you ***must*** use quotation marks and a reference to the source in (Name, Year: Page) format to make clear that you are citing verbatim. Failure to do so may result in allegations of plagiarism. You do not need to cite when including text from the course lecture notes.

Instructions for typing mathematical notation

There may be several places throughout the exam that require mathematical notation. Keep in mind that you will **NOT** be expected to use an equation editor (e.g., Microsoft Word's equation editor) or a typesetting program (e.g., LaTeX) to type this notation. Rather, all questions can be answered using typical keystrokes from a Roman-alphabet-based keyboard (such as that used in the United Kingdom). The following are detailed instructions for how to type the mathematical notation that you may need for the exam:

- **Superscripts**: Use the "^" symbol
  - Example: X^2 would be typed as X^2
- **Subscripts**: Use the "_" symbol
  - Example: Y_0 would be typed as Y_0
- ***Note***: If you need to use both a subscript and a superscript for the same term, use the above "^" and "_" symbols in any order
  - Example: Y_1^K could be typed either as Y_0^K or Y^K_0
- **Addition, Subtraction, and Conditionals**: Use the "+", "-", and "|" symbols exactly as they appear on the keyboard
- **Multiplication**: Use the "*" symbol or no symbol at all, when appropriate
  - Example: $X \times Y$ would be typed as X*Y or XY
- **Division and Fractions**: Use the "/" symbol
  - Example: $X \div Y$ would be typed as X/Y
  - Example: $\frac{X+Y}{Z}$ would be typed as (X + Y) / Z
    - * Note the grouping of the numerator using parentheses; X + Y / Z is different than (X + Y) / Z because of order of operations
- **Sample Means**: Use "-bar" after the variable name
  - $\overline{Y}$ would be typed as Y-bar

You will not necessarily need all of the notational conventions provided above. Further, if you find that you would like to use a notation that is not mentioned, just do your best and provide additional explanation if necessary.

1. Fundamentals

   (a) Consider a hypothetical experiment with twelve units and two experimental groups: treatment and control. The table below contains the potential outcomes Y_1, Y_0 (for treatment and control, respectively), the treatment indicator D (with a value of 1 indicating that a unit was assigned to treatment and a value of 0 indicating a unit was assigned to control), and one additional covariate, X. Complete the following calculations by hand.

Unit	Y_1	Y_0	D	X
1	36	30	1	-0.362
2	27	22	0	-1.332
3	21	21	1	0.239
4	32	28	0	0.285
5	15	11	1	0.253
6	15	12	0	2.710
7	24	21	0	-0.453
8	37	32	1	-0.142
9	12	8	1	-0.896
10	28	25	0	-1.014
11	30	23	1	-0.735
12	23	18	0	0.616

      i. Calculate the true average treatment effect on the treated (ATT) from the experiment based on the potential outcomes. **[2 marks]**

      ii. Calculate the true average treatment effect (ATE) from the experiment based on the potential outcomes. **[2 marks]**

      iii. List the observed potential outcomes from this hypothetical experiment based on the table. **[2 marks]**

      iv. Calculate the estimated average treatment effect (ATE) based on observed outcomes only. **[2 marks]**

      v. Using potential outcomes, calculate the true treatment effects at the level of the individual unit. **[2 marks]**

(b) Imagine that a civic group wanted to increase voter participation. The group focuses on a neighborhood of 5,000 eligible voters and assigns 2,500 to treatment and 2,500 to control. Those assigned to treatment received a series of phone calls encouraging them to watch a televised debate between two candidates. Those assigned to the control group received no contact. They hope to test the causal effect of watching the televised debate on participating in the election.

i. Describe how the stable unit treatment value assumption (SUTVA) might be violated in the scenario described above. **[2 marks]**

ii. Assuming that SUTVA is satisfied, what could be problematic about simply comparing voting rates among those who watched the debate to those who did not watch the debate? **[2 marks]**

iii. If you were advising the organisation on evaluating the effect of watching the debate, what quantity would you suggest that they estimate and why? **[4 marks]**

(c) In many applications, the estimand of most interest to researchers is a simple difference in expected values of an outcome between units that receive a treatment and units that do not. Let Y be an outcome of interest, and let D be an indicator for treatment uptake, with $D = 1$ indicating that a unit is in the treatment group and $D = 0$ indicating that a unit is in the control group. In other words, the estimand of interest is $E[Y_i|D_i = 1] - E[Y_i|D = 0]$, where i indexes subjects. Using potential outcomes notation, this quantity can be further broken down into various meaningful quantities as follows:

$$E[Y_i|D_i = 1] - E[Y_i|D = 0] = E[Y_{1i}|D_i = 1] - E[Y_{0i}|D_i = 1] + E[Y_{0i}|D_i = 1] - E[Y_{0i}|D_i = 0]$$

Using this equation, explain how randomisation solves the selection problem. **[4 marks]**

(d) Imagine a hypothetical experiment with 5 units and two experimental groups: treatment and control. The table below contains the observed outcomes, Y, and the treatment indicator, D. $D = 1$ indicates that a unit was assigned to the treatment group and $D = 0$ indicates that a unit was assigned to the control group.

Unit	Y	D
1	6	1
2	3	0
3	8	1
4	7	0
5	4	1

Calculate the difference in means between treatment and control. Then consider a null hypothesis test in which the null hypothesis is that the difference between treatment and control is 0 and the alternative hypothesis is that the difference between treatment and control is positive. Calculate the p-value of this test using randomisation inference. **[6 marks]**

(e) Suppose that there is a country that has recently experienced civil war between groups largely defined by ethnic affiliation. The international community is interested in implementing policies and programmes that would help the country recover socially and economically. As part of that effort, a non-governmental organisation develops a series of television commercials whose goal is to promote inter-ethnic tolerance. Later, they measure levels of inter-ethnic tolerance in the society using public opinion surveys. If researchers wanted to determine the causal effect of the television commercials on inter-ethnic tolerance, describe what they would require in order to have a plausible causal identification strategy. **[4 marks]**

2. Selection on Observables

   (a) The two main identification assumptions necessary to interpret a treatment effect estimate from a matching procedure as a causal effect are the selection on observables and common support assumptions. If we wanted to estimate an average treatment effect (ATE) from a matching procedure, the required assumptions are stated formally as follows:

   - $(Y_1, Y_0) \perp\!\!\!\perp D | X$ (selection on observables); and
   - $0 < \Pr(D = 1|X = x) < 1$ (common support),

   where D is a binary treatment indicator, Y_0 and Y_1 are potential outcomes, and X is a set of observed covariates.

   If, on the other hand, we wanted to estimate an average treatment effect on the treated (ATT) from a matching procedure, the required assumptions can be relaxed as follows:

   - $Y_0 \perp\!\!\!\perp D | X$ (selection on observables); and
   - $\Pr(D = 1|X = x) < 1$, with $\Pr(D = 1) > 0$ (common support).

   State the relaxed versions of the of the selection on observables and common support assumptions in words. Further, explain in your own words why we are able to relax these assumptions when we are estimating an ATT as opposed to an ATE. **[6 marks]**

   (b) The propensity score theorem of Rosenbaum and Rubin states that if $0 < \Pr(D = 1|X) < 1$ and $(Y_1, Y_0) \perp\!\!\!\perp D | X$, then $(Y_1, Y_0) \perp\!\!\!\perp \pi(X)$, where D is a treatment indicator, Y_0 and Y_1 are potential outcomes, X are covariates, and $\pi(X)$ is the propensity score. Explain this theorem in your own words. **[4 marks]**

   (c) The most common version of estimating a causal effect from a matching procedure makes an explicit choice to estimate an average treatment effect on the treated rather than an average treatment effect. Explain why estimating an average treatment effect on the treated is the typical choice. **[4 marks]**

(d) Imagine that we want to examine the causal effect of watching a television advertisement on subsequently purchasing the product being advertised. In one situation, we are able to perform an experiment. We bring people into a laboratory, randomly assign them to treatment (watching the commercial) or control (not watching anything), and then later track their consumption behaviour to find out whether they bought the product. In another situation, we gather data on past viewing of the television advertisement and subsequent consumption behaviour. In the experiment, we compare between treatment and control by performing a simple difference of means between groups. In the non-experiment, we compare between treatment and control by constructing a matched sample based on 20 measured covariates and calculating an average treatment effect on the treated.

After constructing the matched sample in the non-experiment, we find that there is balance between treatment and control with respect to all 20 covariates. In the experiment, we conduct balance tests with respect to those same 20 covariates and find that 2 exhibit a lack of balance (using a standard t-test of a difference of means between groups). If we assume that the randomisation procedure was properly conducted, would you have more confidence in the estimate of the causal effect based on the experimental data or the non-experimental data? Explain your answer. **[6 marks]**

(e) In various matching procedures, different choices that can be made lead to an explicit tradeoff between bias and variance. Discuss the bias-variance tradeoff in the context of 1:1 as opposed to 1:M distance-based matching, as well as in the context of matching with replacement as opposed to matching without replacement. **[6 marks]**

3. Selection on Unobservables

   (a) A key identification assumption of sharp regression discontinuity designs is the continuity of the relationship between the forcing variable and the outcome around the threshold that assigns units to treatment or control. In other words, in a sharp regression discontinuity design with potential outcomes Y_0 and Y_1, forcing variable X, sharp threshold $X = c$, and binary treatment indicator D, with $D = 1$ if $X \leq c$ and 0 otherwise, the identification assumption is that $E[Y_0|X, D]$ ad $E[Y_1|X, D]$ are continuous in X around $X = c$.

      i. One way this primary identification assumption may be violated is if there is so-called sorting around the threshold. Describe what sorting around the threshold means and describe one way that you might indirectly test for sorting. **[6 marks]**

      ii. Another common indirect test for the validity of the identification assumption is to check for balance in observed covariates between the treatment and control groups. Describe the logic of examining balance. **[4 marks]**

   (b) Describe the parallel trends assumption of difference-in-differences designs in your own words. **[2 marks]**

   (c) If the parallel trends assumption holds, what causal estimand is estimated by performing the difference-in-differences calculation? **[2 marks]**

   (d) The difference-in-differences estimate is produced by using measurements of the outcome variable from two time periods: one pre-intervention and one post-intervention. Given this, would it be useful to have outcome measurements from a longer time series? Why or why not? **[2 marks]**

(e) Consider a situation in which an activist group wants to investigate the impact of participating in a survey about policing practices on later protest participation. In a major UK city, a sample of 20,000 people were randomly assigned to either receive encouragement to participate in an online survey or to be in a control group that was not contacted. Two weeks after the online survey was administered, a protest to criticise police practices was organised by the activist group. Participants in the protest met in a public park and signed in so that their participation was recorded. Assume that the group was able to record accurately whether individuals participated in either or both of the online survey and the protest. The following table presents a summary of the data:

	Group size	Proportion who participated in the protest
Assigned to treatment (encouragement)		
Participated in survey	14,220	.85
Did not participate in survey	5,780	.20
Assigned to control (no encouragement)		
Participated in survey	0	0
Did not participate in survey	20,000	.19

Suppose that the group wants to estimate the causal effect of participating in the survey on subsequently participating in the protest in an instrumental variables framework. In such a framework, let Y be the proportion of people who participated in the protest, let D be a binary indicator for whether a subject in the study participated in the survey (yes = 1, no = 0), and let Z be a binary indicator for whether a subject was encouraged to participate in the survey.

i. Using the data in the table, calculate the naïve ordinary least squares estimate of the effect of participating in the survey on later participating in the protest. That is, estimate β_2 in the following regression: $Y = \beta_1 + \beta_2 D + \epsilon$. **[4 marks]**

ii. Using the data in the table, calculate the intention-to-treat (ITT) effect of receiving encouragement to participate in the survey on subsequent participation in the protest. That is, estimate γ_2 in the following regression: $Y = \gamma_1 + \gamma_2 Z + \epsilon$. **[4 marks]**

iii. Using the data in the table, calculate the proportion of compliers in the study. **[4 marks]**

iv. Using the data in the table, calculate the instrumental variables estimate of the local average treatment effect (LATE) for the compliers. **[4 marks]**

v. Using the data in the table, calculate the average treatment effect on the treated. **[2 marks]**

vi. Formally state the identification assumptions that must be satisfied in order for an instrumental variables procedure to identify a causal effect for compliers. Explain each of these in your own words **[8 marks]**